Ahmed Mohamed Abdulsalam

AI Engineer

LinkedIn GitHub Kaggle Portfolio

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Military Status: Completed

AI Engineer and Computer Science graduate with expertise in deep learning, computer vision, and classical machine learning. Skilled in building production-ready AI applications using Python, TensorFlow, PyTorch, and scikit-learn with deployment via FastAPI and Docker. Experienced across object detection, image classification, OCR, NLP, and predictive modeling with proven track record of published packages and measurable results.

Education

Bachelor's degree at Computer Science, Suez Canal University, Ismailia, September 2024.

Grade: Very Good

Experience

Freelance Machine Learning Engineer, Remote, January 2023 - Present.

- Developed AI-powered applications using TensorFlow and scikit-learn for food recognition and automotive price prediction.
- Built Calories Image Detector: FastAPI application classifying 35 food categories with calorie estimation using CNN model.
- Created Car Price Prediction API: Gradient Boosting REST API with Docker deployment and interactive documentation.

Skills

Core: Python, OpenCV, TensorFlow, PyTorch, scikit-learn, YOLOv8, Tesseract OCR.

Computer Vision: Object detection, image classification, OCR preprocessing, CNN architectures, image preprocessing, SAHI.

Deployment: FastAPI, Docker, ONNX, OpenVINO, REST API design, PyPI packaging, Git.

Data & Tools: NumPy, Pandas, Matplotlib, Seaborn, Jupyter Notebook.

Additional: NLP, LangChain, RAG pipelines, reinforcement learning.

Projects

Screw Detector

- Tiny object detection system using YOLOv8 and SAHI achieving 93.3% F1-score. Implemented slicing-based inference for high-precision detection of small objects with production deployment via ONNX and OpenVINO.

FitSync (Graduation Project)

- AI-powered fitness platform with FastAPI backend featuring K-Means clustering (Silhouette: 0.71) for personalized workout and nutrition recommendations. Built ML models: Random Forest calorie prediction (R^2: 91.7%), SVM heart rate classification (95.9% accuracy), and health-based user/food clustering. Developed cross-platform ecosystem with mobile app, web dashboard, and REST API serving 10+ health and fitness endpoints.

RobustDocOCR

- Robust preprocessing pipeline for document OCR improving Tesseract accuracy by 12%. Implemented adaptive thresholding, Hough transform deskewing, and morphological denoising. Published as PyPI package with CLI and comprehensive documentation.

Mood Detection

- CNN-based deep learning application for real-time facial expression classification (Happy/Sad) achieving 81% accuracy. Built with custom architecture, data augmentation, and GUI supporting both camera input and file upload.

Soundify - Arabic OCR & Text-to-Speech

- Desktop application integrating Tesseract OCR and custom CNN model for Arabic text extraction with text-to-speech synthesis. Features dual recognition modes, real-time camera capture, and modern GUI with dark/light themes.

CBIR - Content-Based Image Retrieval

- Content-based image retrieval system using convolutional autoencoder on CIFAR-10. Implemented latent space embeddings for similarity search with Euclidean distance and integrated image denoising functionality with mAP evaluation.

Courses

Supervised Machine Learning: Regression and Classification from Coursera, February 2024.

Python Project: pillow, tesseract, and OpenCV from Coursera, March 2024.

Volunteering

Chairman at Mish Hackers, Ismailia, January 2020 - December 2024.

Head of Operations at GDSC, Ismailia, January 2022 - January 2023.

Head of Operations at Nasa Space Apps, Ismailia, January 2021 - January 2022.

Languages

Arabic – Native.

English – Intermediate.